The Structure Beneath

How Geometry and Graphs Shape What Models Can Learn

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Sabbatical Talk at ANU

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Introduction

About Me

Dongwoo Kim

Associate Professor, POSTECH

CSE & Graduate School of AI

- PhD, KAIST (2015) — Topic Models, NLP
- Postdoc & Lecturer, ANU (2015–2019)
- Assistant → Associate Prof, POSTECH (2019–)
- Honorary Lecturer, ANU (2019–2024)
- Visiting Researcher, ANU (2026)

Photo

POSTECH Machine Learning Lab

PhD & Combined Students

- Moonjeong Park – GNNs, Dynamical Systems
- Seungbeom Lee – Geometric DL, ML for Bio
- Seunghyuk Cho – Generative, Riemannian
- Jaeseung Heo – GNNs, Data-Centric AI
- Saemi Moon – Trustworthy AI
- Youngbin Choi – LLM Personalization
- Minjong Lee – Generative Models

PhD Students (cont.)

- Kyeongheung Yun – GNNs
- Seokwon Yoon – Machine Learning
- Chongtae Ahn – ML (Industrial, POSCO)

MS Students

- Yesong Ko – Machine Learning
- Sunghyun Choi – ML, Generative Models

A Long History with ANU

2015–2019: Postdoc & Lecturer at ANU (Advisors: Lexing Xie, Cheng Soon Ong)

Qing Wang — 4 joint papers on GNNs (AAAI '23, ICML '23, ICLR '25, ECML '21)

Zhenyue Qin, Yang Liu, Tom Gedeon — Action recognition, GNNs, VLMs

2026: Sabbatical visit — *It feels like coming home*

The Journey

"How should we represent, learn from, and control structured knowledge?"

Structure is both the solution and the problem.

Graphs encode relations, geometry captures hierarchy, symmetry constrains models —

but structure also introduces over-smoothing, noise propagation, and hidden bias.

Understand
What can structured models learn?

→

Question
When does structure mislead?

→

Control
Can we trust what they learn?

→

Apply
Structure bridges to new domains

$\sim$20 papers at top venues (NeurIPS, ICML, ICLR, ICCV, EMNLP, AAAI, . . .)

Act 1: What Can Structure-Aware Models Learn?

Breaking the Expressivity Barrier

The challenge: Standard message-passing GNNs are bounded by the 1-WL test

Key idea: Use local vertex colouring to break symmetry

- Assign initial colours based on local structure
- GNNs with vertex colouring distinguish graphs beyond 1-WL
- Higher expressivity *without* expensive higher-order message passing

Result: More expressive GNNs that remain practically efficient

[Figure: Vertex colouring illustration]

When Structure Hinders: Heterophilic Graphs

Shouheng Li, Dongwoo Kim, Qing Wang **AAAI 2023** 🤝 ANU Collab

The problem: GNNs assume neighbours are similar — but what if they're not?

Heterophilic graphs violate the homophily assumption underlying most GNNs

Key idea: Don't fight the graph — *restructure* it

- Learnable spectral clustering rewires edges
- Increases effective homophily ratio
- GNN then operates on a friendlier graph topology

Result: Significant gains on heterophilic benchmarks

[Figure: Graph restructuring pipeline]

Bridging Expressivity and Generalization

Shouheng Li, Floris Geerts, Dongwoo Kim, Qing Wang

ICLR 2025 🤝 ANU Collab

Core question: Is there a fundamental trade-off between expressivity and generalization?

Key contributions:

- Unified framework connecting WL hierarchy (expressivity) with Rademacher complexity (generalization)
- Higher expressivity need *not* hurt generalization
- Practical guidelines for principled GNN architecture design

In loving memory of our collaborator and friend, Qing Wang.

Four papers together — this line of work is her legacy too.

Theory says GNNs are powerful —

but practitioners see degradation in deep networks.

What's going wrong?

Act 2: When Structure Misleads

Turning Smoothing into a Feature

MoonJeong Park, Jaeseung Heo, Dongwoo Kim

Insight: Message passing is a diffusion process — can we *reverse* it?

- Standard GNNs: forward diffusion $\rightarrow$ smoothing $\rightarrow$ information loss
- Our approach: learn a *reverse* diffusion process
- Recovers discriminative node features even on heterophilic graphs
- Principled framework connecting diffusion models with GNNs

Result: State-of-the-art on heterophilic benchmarks while maintaining homophilic performance

[Figure: Forward vs. reverse diffusion on graphs]

Working With Structure, Not Against It

Jaeseung Heo, Moonjeong Park, Dongwoo Kim

AAAI 2026 Oral

Instead of reversing smoothing — what if we smooth smarter?

Key idea: Posterior label smoothing for node classification

- Smooth labels using graph-structure-aware posterior distribution
- Adapts smoothing strength based on local neighbourhood
- Theoretically principled: connects to Bayesian label propagation

Result: Consistent improvements across homophilic *and* heterophilic graphs

From fighting structure to collaborating with it — an evolving perspective.

The Oversmoothing Fallacy

MoonJeong Park, Sunghyun Choi, Jaeseung Heo, Eunhyeok Park, Dongwoo Kim arXiv

Provocative claim: Over-smoothing may be a misguided narrative

Careful empirical study reveals:

- Performance degradation in deep GNNs is *not* always due to over-smoothing
- Optimization issues (vanishing gradients, training instability) are often the real culprit
- Standard metrics for over-smoothing can be misleading

The arc: Reverse it (ICML'24) → Work with it (AAAI'26) → Maybe it's not even the problem

Sometimes the most important contribution is questioning the question itself.

The Model Is Only as Good as Its Data

If imperfect structure corrupts learning, what about imperfect data?

Jaeseung Heo*, Seungbeom Lee*, Sungsoo Ahn, Dongwoo Kim **IJCAI 2024 EPIC: Graph Augmentation via Edit Path Interpolation**

- Define augmentation via edit path interpolation between graphs
- Generates structurally diverse yet semantically consistent graphs

Jaeseung Heo, Kyeongheung Yun, Seokwon Yoon, MoonJeong Park, Jungseul Ok, Dongwoo Kim **NeurIPS 2025 Influence Functions for Edge Edits in GNNs**

- Extend influence functions to edge-level edits in non-convex GNNs
- Which edges matter most? Efficiently approximate the effect of adding/removing edges

When Labels Lie: Noise Propagation in Graphs

Suyeon Kim, SeongKu Kang, Dongwoo Kim, Jungseul Ok, Hwanjo Yu

KDD 2025

Gap: Instance-dependent label noise is well-studied for images, not for graphs

- Noise patterns in graphs are fundamentally different from i.i.d. data
- Graph structure both *propagates* and *helps correct* label noise
- Comprehensive benchmark for evaluating robustness to label noise in GNNs

Shared insight across all three works:

Augment what's missing (EPIC), measure what matters (influence), and understand how errors spread (label noise). *Data quality is a structural problem.*

If imperfect data corrupts learning,

what happens when we *need* to remove data?

And how do we verify our models are actually robust?

Act 3: Can We Trust What Structured Models Learn?

Making Models Forget: From Data Points to Features

The same team, an increasingly ambitious question

Youngsik Yoon*, Jinhwan Nam*, Jaeho Lee, Dongwoo Kim, Jungseul Ok **IEEE S&P 2024 Few-shot Unlearning**

- Remove data influence with access to only a few samples
- Provable guarantees on forgetting quality
- Published at S&P — bridging ML and security

Saemi Moon, Seunghyuk Cho, Dongwoo Kim **AAAI 2024 Feature Unlearning for Generative Models**

- Go beyond data points: unlearn features from GANs and VAEs
- E.g., remove ability to generate glasses on faces
- Feature-level unlearning is more natural for generative models

From Features to Concepts: Unlearning at Scale

Saemi Moon*, Minjong Lee*, Sangdon Park, Dongwoo Kim

ICCV 2025

How do we properly evaluate unlearning for text-to-image models?

HUB: Holistic Unlearning Benchmark

- Evaluates along multiple axes:
  - Forgetting quality — is the concept truly removed?
  - Generation quality — does the model still work well?
  - Robustness — can unlearning be circumvented?
- Reveals surprising failure modes in existing methods

The unlearning arc: Data points (S&P'24) → Features (AAAI'24) → Concepts & evaluation (ICCV'25)

Each step raised the stakes — and the bar for verification.

Rigorous Evaluation as a Contribution

Minjong Lee, Dongwoo Kim

ICCV 2023 Oral

Context: Diffusion-based purification claimed strong adversarial robustness

Our finding: Previous evaluations were fundamentally flawed

- Gradient-based attacks fail due to stochastic purification process
- We design proper evaluation methodology with stronger attack protocols
- Robustness claims were significantly overstated

Lesson: Rigorous evaluation is as important as strong defences

Selected as Oral at ICCV 2023 — the community recognized that *proving something doesn't work* can be the most valuable contribution.

Bias Hides in Structure Too

Fair Music Recommendation

Jinhyeok Park*, Dain Kim*, Dongwoo Kim

CIKM 2022 Best Paper

- Item-based VAE for fair recommendation
- Mitigates popularity bias while maintaining quality
- Won Best Paper at EvalRS Data Challenge

ChronoBias for LLMs

Kyungmin Kim, Youngbin Choi, Hyounghun Kim, Dongwoo Kim, Sangdon Park

EMNLP Findings 2025

- Benchmark for time-conditional group bias
- LLMs reflect biases from specific time periods
- First systematic study of temporal bias

Bias is structural: it lives in recommendation graphs and temporal data distributions alike.

The principles we've developed —

respect geometry, question assumptions, verify rigorously —

apply far beyond graphs.

Act 4: Structure as Bridge to New Domains

Matching Geometry to Data: Hyperbolic Representations

When Euclidean space isn't enough: Hierarchical data needs hyperbolic geometry

Seunghyuk Cho, Juyong Lee, Jaesik Park, Dongwoo Kim **NeurIPS 2022 Rotated Hyperbolic Wrapped Normal**

- Standard wrapped normal is isotropic — too rigid
- Introduce rotation for anisotropic distributions on the Lorentz model

Seunghyuk Cho, Juyong Lee, Dongwoo Kim **NeurIPS 2023 Hyperbolic VAE via Latent Gaussian Distributions**

- **Problem → clean solution:** Hyperbolic VAEs are numerically unstable
- Work in tangent space with Gaussians, map via exponential map
- Stable training with state-of-the-art hierarchical generation

Drug Design with Multi-Reward Optimization

Seungbeom Lee*, Munsun Jo*, Jungseul Ok, Dongwoo Kim

ICML 2025

Problem: Drug design requires optimizing multiple competing objectives

- Ligand validity, binding affinity, drug-likeness — all matter simultaneously
- Single-reward optimization leads to degenerate solutions
- **Our approach:** Multi-reward optimization for structure-based drug design
- Molecular graphs as the natural representation
- Pareto-optimal generation balances competing rewards

Result: More realistic drug candidates with better binding properties

Physical Symmetry Meets Generative Modelling

Seongsu Kim, Nayoung Kim, Dongwoo Kim, Sungsoo Ahn **NeurIPS 2025** **Spotlight**

Context: DFT is fundamental to computational chemistry but prohibitively expensive

- High-order equivariant flow matching for Hamiltonian prediction
- Respects physical symmetries: rotation, translation, permutation
- Flow matching provides efficient and stable training
- Orders of magnitude faster than direct DFT computation

Impact: Accelerating quantum chemistry with geometric ML

Selected as **Spotlight** at NeurIPS 2025 — when the geometry is right, the physics follows.

Collaborative Structure for LLM Personalization

Youngbin Choi, Seunghyuk Cho, Minjong Lee, MoonJeong Park, Yesong Ko, Jungseul Ok, Dongwoo Kim

EMNLP 2025

Problem: LLM alignment treats all users the same — but preferences differ

CoPL: Collaborative Preference Learning

- Collaborative approach to personalizing LLMs
- Learn shared preference structure across users
- Individual personalization through collaborative filtering on preferences
- Efficient: no per-user fine-tuning needed

Insight: User preferences have *structure* — collaborative filtering exploits it, just as GNNs exploit graph structure and equivariant models exploit symmetry.

Geometric Reasoning: An ANU Collaboration

Seunghyuk Cho, Zhenyue Qin, Yang Liu, Youngbin Choi, Seungbeom Lee, Dongwoo Kim **EMNLP Findings 2025** 🤝 ANU Collab **GeoDANO: Geometric VLM with Domain-Agnostic Vision Encoder**

- Vision-language models struggle with geometric reasoning
- Domain-agnostic encoder: not tied to natural images, diagrams, or plots
- Strong geometric reasoning without domain-specific pre-training

Seunghyuk Cho, Zhenyue Qin, Yang Liu, Youngbin Choi, Seungbeom Lee, Dongwoo Kim **EACL Findings 2026** 🤝 ANU Collab **Plane Geometry Problem Solving: A Survey**

- Comprehensive survey: parsing, reasoning, symbolic & neural approaches
- Roadmap for the community on geometric reasoning

Collaborative work with **Zhenyue Qin** and **Yang Liu** (ANU) — the connection continues.

What Structure Taught Us

Recurring Lessons

Respect Geometry

Match representation to data:
hyperbolic for hierarchies,
equivariant for physics,
graphs for relations

? Question Narratives

Over-smoothing might not be
the real problem.
Robustness claims need
rigorous evaluation.

Verify Claims

Build benchmarks (HUB).
Design proper evaluation.
The hardest contribution
is proving something wrong

These principles — developed across GNNs, unlearning, and scientific ML —
define how our lab approaches *every* new problem.

Open Questions

Structure at Scale

Can we scale structural inductive biases to foundation models?

Or will scale make structure obsolete?

Trustworthy GenAI

Unlearning meets billion-parameter models.

Can we verify forgetting when we can't even verify generation?

Scientific Discovery

Will geometric ML transform science, or just accelerate it?

The difference matters.

Continuing the ANU connection

GNN theory (Qing's legacy) · Geometric reasoning · Student exchange & co-supervision

Thank you!

Questions?

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Backup: Publication Timeline (2025–2026)

Year	Venue	Topic
2026	AAAI (Oral)	Posterior Label Smoothing
2026	EACL Findings $\times 2$	Geometry Survey, Controllable Summ.
2025	NeurIPS (Spotlight)	Equivariant Flow Matching for DFT
2025	NeurIPS	Influence Functions for Edge Edits
2025	EMNLP	CoPL: Collaborative Preference Learning
2025	EMNLP Findings $\times 2$	GeoDANO, ChronoBias
2025	ICCV	Holistic Unlearning Benchmark
2025	KDD	Label Noise in Graph Data
2025	ICML	Drug Design Multi-Reward
2025	ICLR	Bridging Generalization & Expressivity

Backup: Publication Timeline (2022–2024)

Year	Venue	Topic
2024	ICML	Reverse Process GNN
2024	IJCAI	EPIC: Graph Augmentation
2024	IEEE S&P	Few-shot Unlearning
2024	AAAI	Feature Unlearning
2023	NeurIPS	Hyperbolic VAE
2023	ICCV (Oral)	Robust Eval of Diffusion Purification
2023	ICML	Local Vertex Colouring GNN
2023	AAAI $\times 2$	Restructuring Graph, Skeleton Anon.
2022	NeurIPS	Rotated Hyperbolic Wrapped Normal
2022	CIKM (Best Paper)	Fair Music Recommendation
2022	AISTATS	Robust Learning from Crowds

Backup: ANU Collaborations

- **Qing Wang** (ANU $\rightarrow$ in memoriam)
  - Beyond Low-Pass Filters (ECML-PKDD 2021)
  - Restructuring Graph for Higher Homophily (AAAI 2023)
  - Local Vertex Colouring GNN (ICML 2023)
  - Bridging Generalization and Expressivity (ICLR 2025)
- **Zhenyue Qin, Yang Liu, Tom Gedeon** (ANU)
  - Anonymization for Skeleton Action Recognition (AAAI 2023)
  - Position-Sensing GNN (TNNLS 2024)
  - Fusing Higher-Order Features for Action Recognition (TNNLS 2022)
  - GeoDANO: Geometric VLM (EMNLP Findings 2025)
  - Geometry Problem Solving Survey (EACL Findings 2026)